Tasca Avaluable 7.4

1. Carrega la informació dels arxius Transactions.json i Accounts.json al mysql workbench en dues taules. Explica tot el procés amb captures de pantalla i copia el codi.

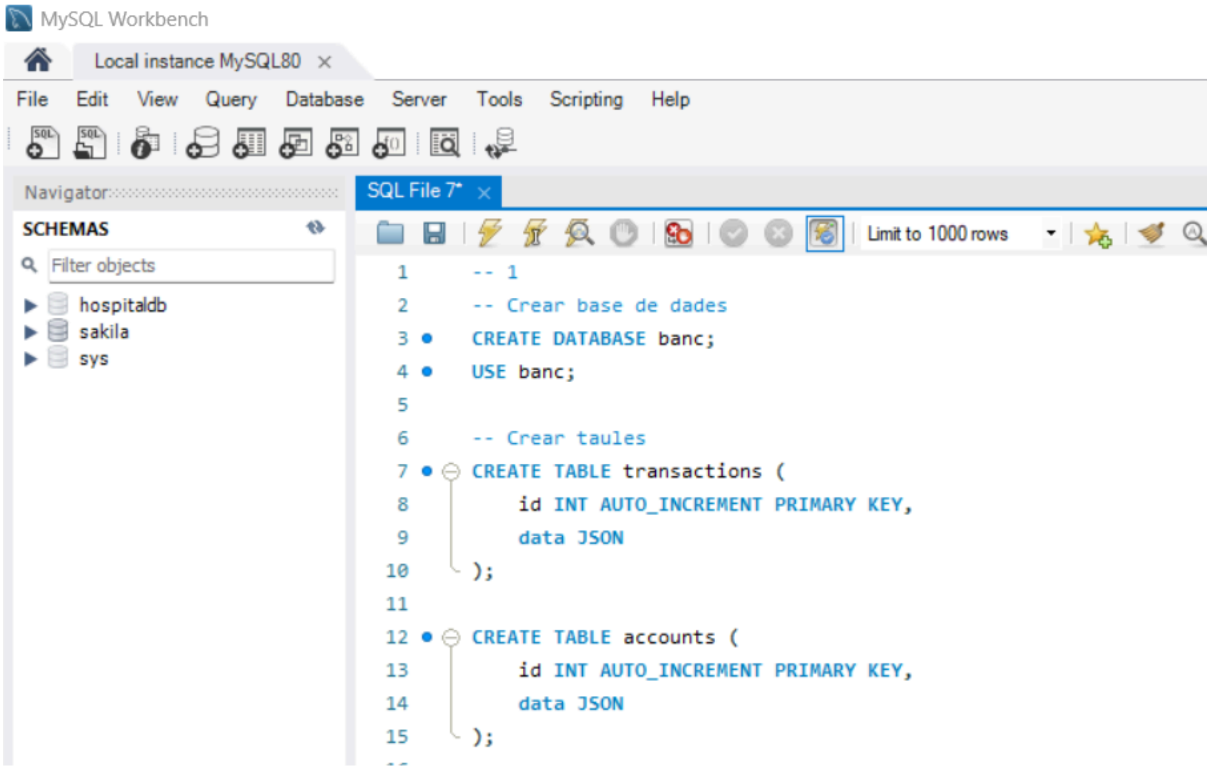
```
-- 1
-- Crear base de dades
CREATE DATABASE banc;
USE banc;

-- Crear taules
CREATE TABLE transactions (
   id INT AUTO_INCREMENT PRIMARY KEY,
   data JSON
);

CREATE TABLE accounts (
   id INT AUTO_INCREMENT PRIMARY KEY,
   data JSON
);

-- Importar les dades
INSERT INTO transactions (data)
VALUES (
   CONVERT(
      LOAD_FILE('C:/ProgramData/MySQL/MySQL Server 8.0/Uploads/Transactions.json')
      USING utf8mb4
   )
);

INSERT INTO accounts (data)
VALUES (
   CONVERT(
      LOAD_FILE('C:/ProgramData/MySQL/MySQL Server 8.0/Uploads/Accounts.json')
      USING utf8mb4
   )
);
```


MySQL Workbench
Local instance MySQL80
File Edit View Query Database Server Tools Scripting Help
Navigator
SCHEMAS
Filter objects
hospitaldb
sakila
sys
SQL File 7*
Limit to 1000 rows
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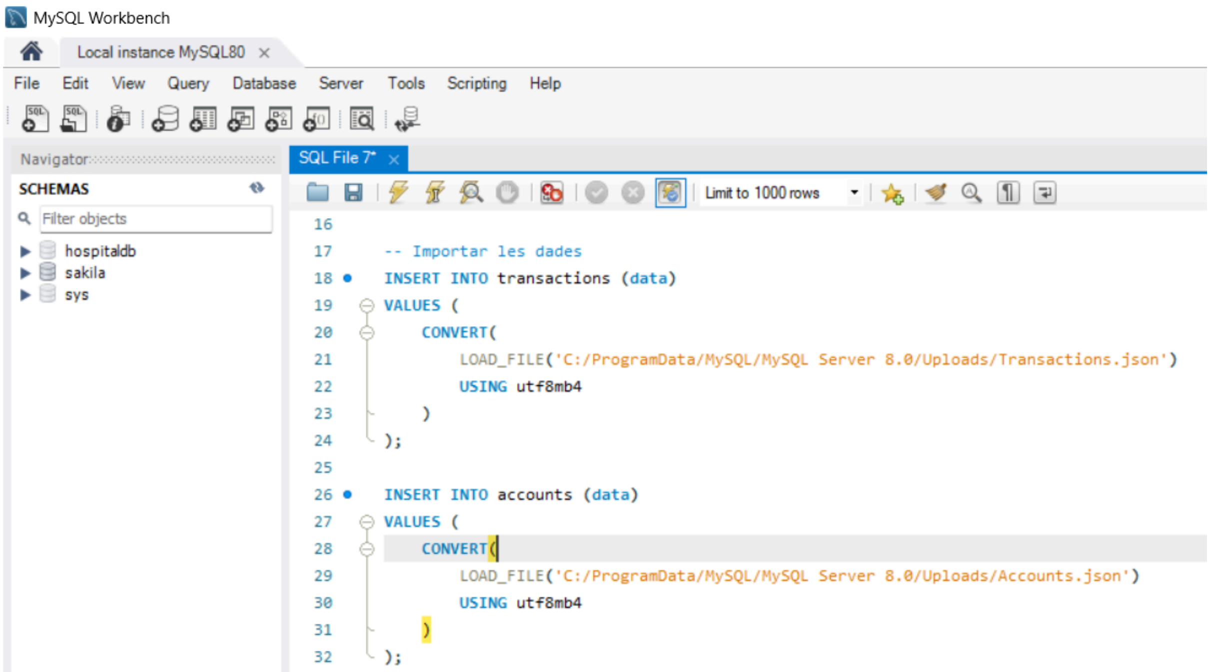
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VALUES (
CONVERT(
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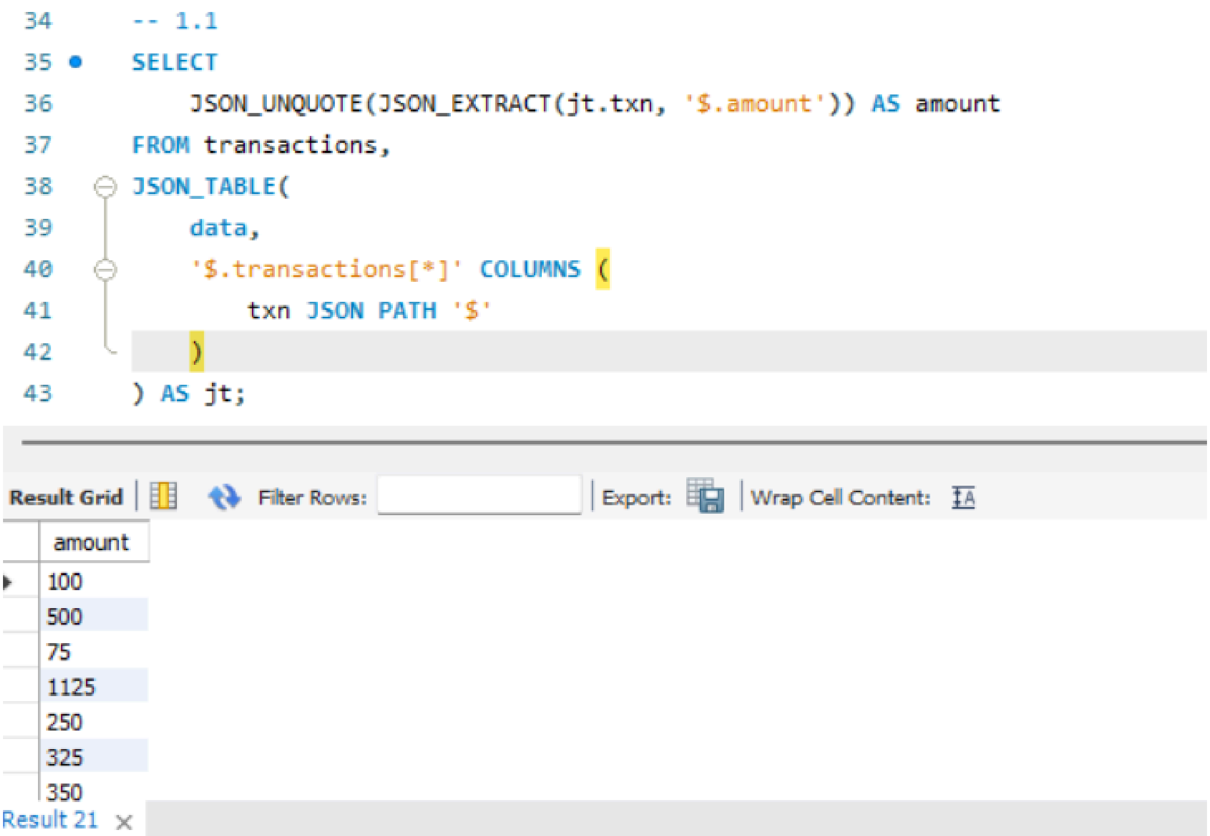
USING utf8mb4
)
);
INSERT INTO accounts (data)
VALUES (
CONVERT(
LOAD_FILE('C:/ProgramData/MySQL/MySQL Server 8.0/Uploads/Accounts.json')
USING utf8mb4
)
);

1.1. Quina consulta utilitzaries per obtenir l'import (amount) de totes les transaccions de la taula transactions (Utilitza la funció JSON_UNQUOTE per a eleminar les cometes)?

```
SELECT
   JSON_UNQUOTE(JSON_EXTRACT(jt.txn, '$.amount')) AS amount
FROM transactions,
JSON_TABLE(
   data,
   '$.transactions[*]' COLUMNS (
      txn JSON PATH '$'
   )
) AS jt;
```



```
-- 1.1
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    JSON_UNQUOTE(JSON_EXTRACT(jt.txn, '$.amount')) AS amount
FROM transactions,
JSON_TABLE(
    data,
    '$.transactions[*]' COLUMNS (
        txn JSON PATH '$'
    )
) AS jt;
```


amount
100
500
75
1125
250
325
350

1.2. Com podries filtrar les transaccions que tenen una quantitat superior a 200 (utilitza JSON_EXTRACT?

```
SELECT
   JSON_UNQUOTE(JSON_EXTRACT(jt.txn, '$.txn_id')) AS txn_id,
   JSON_OBJECT(
      'amount', CAST(JSON_EXTRACT(jt.txn, '$.amount') AS UNSIGNED),
      'txn_date', JSON_UNQUOTE(JSON_EXTRACT(jt.txn, '$.txn_date'))
   ) AS txn_data
FROM transactions,
JSON_TABLE(
   data,
   '$.transactions[*]' COLUMNS (
      txn JSON PATH '$'
   )
) AS jt
WHERE CAST(JSON_EXTRACT(jt.txn, '$.amount') AS DECIMAL(10,2)) > 200;
```

```
-- 1.2
SELECT
    JSON_UNQUOTE(JSON_EXTRACT(jt.txn, '$.txn_id')) AS txn_id,
    JSON_OBJECT(
        'amount', CAST(JSON_EXTRACT(jt.txn, '$.amount') AS UNSIGNED),
        'txn_date', JSON_UNQUOTE(JSON_EXTRACT(jt.txn, '$.txn_date'))
    ) AS txn_data
FROM transactions,
JSON_TABLE(
    data,
    '$.transactions[*]' COLUMNS (
        txn JSON PATH '$'
    )
) AS jt
WHERE CAST(JSON_EXTRACT(jt.txn, '$.amount') AS DECIMAL(10,2)) > 200;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

txn_id	txn_data
2	{"amount": 500, "txn_date": "2000-01-15 00:0...
4	{"amount": 1125, "txn_date": "2001-03-12 00:...
5	{"amount": 250, "txn_date": "2001-03-12 00:0...
6	{"amount": 325, "txn_date": "2002-11-23 00:0...
7	{"amount": 350, "txn_date": "2002-12-15 00:0...
10	{"amount": 452, "txn_date": "2004-09-30 00:0...

1.3. Escriu una consulta que obtingui l'ID de la transacció (txn_id) i la data de la transacció (txn_date) per totes les transaccions.

```
SELECT
   JSON_UNQUOTE(JSON_EXTRACT(jt.txn, '$.txn_id')) AS txn_id,
   JSON_UNQUOTE(JSON_EXTRACT(jt.txn, '$.txn_date')) AS txn_date
FROM transactions,
JSON_TABLE(
   data,
   '$.transactions[*]' COLUMNS (
      txn JSON PATH '$'
   )
) AS jt
ORDER BY CAST(JSON_UNQUOTE(JSON_EXTRACT(jt.txn, '$.txn_id')) AS UNSIGNED);
```

```
-- 1.3
SELECT
    JSON_UNQUOTE(JSON_EXTRACT(jt.txn, '$.txn_id')) AS txn_id,
    JSON_UNQUOTE(JSON_EXTRACT(jt.txn, '$.txn_date')) AS txn_date
FROM transactions,
JSON_TABLE(
    data,
    '$.transactions[*]' COLUMNS (
        txn JSON PATH '$'
    )
) AS jt
ORDER BY CAST(JSON_UNQUOTE(JSON_EXTRACT(jt.txn, '$.txn_id')) AS UNSIGNED);
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

txn_id	txn_date
1	2000-01-15 00:00:00
2	2000-01-15 00:00:00
3	2004-06-30 00:00:00
4	2001-03-12 00:00:00
5	2001-03-12 00:00:00
6	2002-11-23 00:00:00
7	2002-12-15 00:00:00

1.4. Escriu una consulta que recuperi l'ID del compte (account_id) i l'import (amount) de les transaccions, unint les taules transactions i accounts per account_id.

SELECT
  CAST(JSON_UNQUOTE(JSON_EXTRACT(t.txn, '$.account_id')) AS UNSIGNED) AS account_id,
  CAST(JSON_EXTRACT(t.txn, '$.amount') AS DECIMAL(10,2)) AS amount
FROM transactions AS tr
CROSS JOIN JSON_TABLE(
  tr.data,
  '$.transactions[*]' COLUMNS (
    txn JSON PATH '$'
)
) AS t
JOIN accounts AS ac
CROSS JOIN JSON_TABLE(
  ac.data,
  '$.accounts[*]' COLUMNS (
    acc JSON PATH '$'
)
) AS a
  ON CAST(JSON_UNQUOTE(JSON_EXTRACT(t.txn, '$.account_id')) AS UNSIGNED) =
    CAST(JSON_UNQUOTE(JSON_EXTRACT(a.acc, '$.account_id')) AS UNSIGNED)
ORDER BY account_id;

```
-- 1.4
SELECT
    CAST(JSON_UNQUOTE(JSON_EXTRACT(t.txn, '$.account_id')) AS UNSIGNED) AS account_id,
    CAST(JSON_EXTRACT(t.txn, '$.amount') AS DECIMAL(10,2)) AS amount
FROM transactions AS tr
CROSS JOIN JSON_TABLE(
    tr.data,
    '$.transactions[*]' COLUMNS (
        txn JSON PATH '$'
    )
) AS t
JOIN accounts AS ac
CROSS JOIN JSON_TABLE(
    ac.data,
    '$.accounts[*]' COLUMNS (
        acc JSON PATH '$'
    )
) AS a
    ON CAST(JSON_UNQUOTE(JSON_EXTRACT(t.txn, '$.account_id')) AS UNSIGNED) =
       CAST(JSON_UNQUOTE(JSON_EXTRACT(a.acc, '$.account_id')) AS UNSIGNED)
ORDER BY account_id;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

account_id	amount
1	100.00
2	500.00
3	75.00
4	1125.00
5	250.00
7	325.00

1.5. Com podries calcular l'import total de les transaccions per cada compte (account_id)?

```
SELECT
   CAST(JSON_UNQUOTE(JSON_EXTRACT(jt.txn, '$.account_id')) AS UNSIGNED) AS
account_id,
   SUM(CAST(JSON_EXTRACT(jt.txn, '$.amount') AS DECIMAL(10,2))) AS total_amount
FROM transactions AS tr
CROSS JOIN JSON_TABLE(
   tr.data,
   '$.transactions[*]' COLUMNS (
      txn JSON PATH '$'
   )
) AS jt
GROUP BY CAST(JSON_UNQUOTE(JSON_EXTRACT(jt.txn, '$.account_id')) AS
UNSIGNED)
ORDER BY account_id;
```

```
-- 1.5
SELECT
    CAST(JSON_UNQUOTE(JSON_EXTRACT(jt.txn, '$.account_id')) AS UNSIGNED) AS account_id,
    SUM(CAST(JSON_EXTRACT(jt.txn, '$.amount') AS DECIMAL(10,2))) AS total_amount
FROM transactions AS tr
CROSS JOIN JSON_TABLE(
    tr.data,
    '$.transactions[*]' COLUMNS (
        txn JSON PATH '$'
    )
) AS jt
GROUP BY CAST(JSON_UNQUOTE(JSON_EXTRACT(jt.txn, '$.account_id')) AS UNSIGNED)
ORDER BY account_id;
```


account_id	total_amount
1	100.00
2	500.00
3	75.00
4	1125.00
5	250.00
7	325.00
8	350.00

Enllac Github: https://github.com/yanalexanderadames-gif/Llenguatge_De_Marques